MIRO2026 WORKSHOP

November 17 – 20

UNIVERSIDAD DE CONCEPCIÓN

Santiago campus

BOOK OF ABSTRACTS

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Stephen Walborn | Marcel Clerc

Organized by:

MIRO Millennium Institute for Research in Optics

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Universidad de los Andes

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MIRO2026
WORKSHOP

PROGRAM

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08:45-09:00	**Opening**			
09:00-09:15	Cristóbal Lledó	Johan Triana	Darío Pérez	Álvaro Alarcón
09:15-09:30				
09:30-09:45	Joaquín Toloza	Edgar Barriga	Alberto Navarrete	Adrián Rubio
09:45-10:00	Kei Sawada	Luciano Toro	Romina Abarca	Rubén Fritz
10:00-10:15	Katy Aruachan	Daniel Uzcategui	Samir Das	Polette Parra
10:15-10:30	**Coffee Break**			
10:30-10:45				
10:45-11:00	Rodrigo Vicencio	Stephen Walborn	Aldo Delgado	Felipe Herrera
11:00-11:15				
11:15-11:30	Daniel Concha	Eloísa Pérez	María Paz Barria	Hans Mättig
11:30-11:45	Felipe Recabal	Nikolas Naranjo	Nelson Villalba	Manuel Diaz
11:45-12:00	Martin Rubio	Carlos Pirela	Priscila Cáces	Jean Cortés
12:00-13:30	**Lunch Break**			
13:30-13:45	Birger Seifert	Marcel Clerc	Mauricio Morel	Diego Guzmán
13:45-14:00				
14:00-14:15	Poster Preview	Andrés Seguel	Lucas Hernández	Osmar Aravena
14:15-14:30		Gastón González	Belén Orellana	Diego Hidalgo
14:30-14:45	**Coffee Break**			**Coffee Break**
14:45-15:00				**Departure**
15:00-15:15	Poster Session	Discussion Panel: New Frontiers in Science and Technology	Bastián Real	
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15:30-15:45			Santiago Ocampo	
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16:00-16:15			Sofia Pinto	
16:15-16:30				

TALKS

Cristóbal Lledó, Universidad de Chile

Cat qubits in circuit QED

Quantum error correction exploits the fact that most environments are only locally correlated, so information stored nonlocally resists decoherence. Encoding a single logical qubit in many physical qubits, however, makes fault-tolerant quantum computing hardware-expensive. A more efficient alternative is to encode the qubit nonlocally in the phase space of a single oscillator, providing a first layer of protection against local noise.

The cat qubit is a prominent continuous-variable qubit, realized in superconducting circuits through stabilization. It is called a “noise-biased qubit” because it is exponentially robust against bit flips but not phase flips, a property that simplifies error correction.

In this talk, I will explain how we stabilize, control, and read out the superconducting cat qubit. I will briefly mention our recent work on understanding the Kerr-cat qubit’s current limitations.

Joaquín Toloza, Pontificia Universidad Católica de Chile

Implementation of a Photoconductive-Antenna-Based THz Time-Domain Spectroscopy Platform for Nonlinear Materials Studies

We report the implementation and characterization of a terahertz time-domain spectroscopy (THz-TDS) platform based on photoconductive antennas and femtosecond optical excitation. The system enables coherent generation and detection of pulsed THz radiation, allowing the temporal electric-field waveform to be measured and its frequency-domain spectrum to be obtained. Stable and measurable THz signals have been achieved, demonstrating the successful operation of the experimental setup. The platform is intended for spectroscopic studies and for future investigations of THz generation in nonlinear optical materials. Its implementation establishes a new experimental capability for ultrafast THz research in Chile and provides the basis for systematic studies of material response in the terahertz spectral range.

Kei Braz Sawada, Universidad de los Andes, Chile

Construction of a 3-km classical communication link between Cerro Calán Observatory and Universidad de los Andes

We report the construction of a transmitter at Cerro Calán Observatory and the detection of classical light at Universidad de los Andes, establishing a 3-km free-space optical link in preparation for quantum key distribution (QKD). The transmitter is designed to accommodate a bright 660-nm laser providing classical light for adaptive optics, as well as a 785-nm weak coherent state source for polarization-encoded QKD. It is fully remotely controlled from the receiver over an Internet connection. We describe the design of the transmitter's telescope, its gimbal mount, camera-based pointing feedback, and control systems. Trials with classical light are not only a first step for optical alignment in free-space QKD, but also useful on their own for measurements of atmospheric turbulence, which are scarce in highly-polluted environments such as Santiago. This work establishes infrastructure for subsequent adaptive optics experiments and free-space QKD.

Katy Aruachan, Universidad de Santiago de Chile

Hybrid Atomistic-Parametric Decoherence Model for Molecular Spin Qubits

Solid-state molecular qubits offer significant potential for quantum technologies, but understanding the factors limiting their coherence remains challenging. This study introduces a random Hamiltonian approach combining molecular dynamics simulations and quantum master equations to predict relaxation (T_1) and dephasing (T_2) times in copper porphyrin qubits. Results show that conventional spin-lattice interaction models fail to reproduce experimental measurements accurately. By incorporating magnetic field noise arising from nuclear spins in the crystal lattice, the model achieves quantitative agreement across different magnetic fields. The study also reveals distinct magnetic field dependencies: T_1 scales as $1/B$ due to combined spin-lattice and magnetic noise contributions, while T_2 scales as $1/B^2$, driven by low-frequency magnetic noise. These findings demonstrate the potential of atomistic dynamical methods to understand quantum coherence and model the behavior of molecular spin qubits, supporting the development of future quantum technologies.

Rodrigo Vicencio, Universidad de Chile

Multi-orbital photonics

In this talk, I will present recent results of our group on multi-orbital photonics, including atom-like orbital states and vortices. We will show how we can map an input phase difference into a geometrical orientation of dipolar states and how we can, for example, select the topological properties of a dimerized lattice by choosing a specific orbital state.

Daniel Concha, Universidad de Concepción

Certifying Wigner non-classicality at a single phase-space point

Wigner negativity at a single phase-space point is enough to certify non-classicality, so no reconstruction is needed: only one point where displaced parity is negative beyond shot noise. We give a two-phase protocol that reaches the state only through displaced-parity counts. Complex simultaneous perturbation stochastic approximation, which suits real-valued functions of complex arguments and costs two evaluations per iteration at any mode number, nominates a point. A fresh sample there gives one exact Clopper–Pearson bound. This separation keeps the false-positive rate at its nominal level: reusing the search data would certify the vacuum at a rate of 0.718, while fresh sampling gives 0.050. The protocol certifies Fock, cat and squeezed-cat states in at least 97% of runs, with a median of 22,000–50,000 shots, and beats an equal-budget grid scan from two modes on. We also report implementation pitfalls, each identified and resolved.

Felipe Recabal, Universidad de Santiago de Chile

Bath-induced resonant deviations from quantum Gibbs statistics of strongly interacting oscillators

Secular quantum master equations are widely used to study thermalization of interacting quantum systems that are weakly coupled to thermal reservoirs. For quantum systems with dissipation timescales that are comparable with low-frequency oscillations between energy levels, non-secular contributions to thermalization are expected but their influence remains largely unexplored. By implementing a partial secularization procedure, we demonstrate that the steady-state of two coupled quantum oscillators with local baths at thermal equilibrium, deviates from canonical Gibbs statistics for unequal damping rates. For strongly coupled near-degenerate oscillators, non-secular dissipation pathways establishes stationary population flux between sub-systems in the absence of temperature gradients. This population redistribution is a signature of non-Gibbs statistics and arises from bath-induced coherences between closely-spaced eigenstates. Possible experimental signatures of non-canonical population statistics in available quantum platforms are discussed.

Martín Rubio, Universidad de Chile

Physics-Informed Neural Networks for Experimental Validation of Self-Organization in Chiral Liquid Crystals

Pattern formation and dynamics far from primary instabilities expose a persistent gap between minimal theoretical descriptions valid near onset and experimental observations. Closing this gap is essential for advancing our understanding of complex nonlinear systems. Using physics-informed neural networks, we tackle the inverse problem of parameter identification and initial-condition reconstruction in chiral nematic liquid crystals. By embedding the chiral-anisotropic Ginzburg- Landau equation into the network architecture, we simultaneously recover the order-parameter fields and estimate the critical coefficients directly from polarized-light microscopy data. This framework further enables the extraction of temperature-dependent parameter variations and reveals nontrivial transitions in the underlying dynamics.

Birger Seifert, Pontificia Universidad Católica de Chile

Experimental Investigation of Phase and Energy Fluctuations in a Mid-Infrared Multimode OPA

Nondegenerate optical parametric amplifiers (OPAs) provide tunable mid-infrared radiation. Here, we experimentally investigate a multimode OPA operated with continuous-wave (CW) seeding and characterize its transition towards quantum-optical noise behavior. We measure carrier-envelope phase (CEP) fluctuations and pulse-energy fluctuations as a function of the seeding conditions. The experimental results are analyzed using quantum-optical methods to identify the contribution of the multimode structure to the observed noise properties. In particular, we investigate how the number of participating modes affects phase and energy fluctuations and compare the measured behavior with the expected quantum-optical response. The results provide an experimental characterization of noise evolution in a multimode mid-infrared OPA and reveal the influence of multimode operation on the stability of parametrically generated radiation.

Johan Triana, Universidad Católica del Norte

Light-Driven the quantum nature of light

The generation of light states with sub-Poissonian photon statistics has been explored in atomic and molecular systems coupled to Fabry-Perot cavities or plasmonic nanostructures. Despite significant advances in recent years in quantum technologies, generating and manipulating states of light whose variance of the photon number is smaller than the mean photon number remains a major challenge. We propose a novel approach, based on optimal control theory, to generate nonclassical states of light by modifying the dynamics of molecular systems confined in an infrared electromagnetic environment. We implement the steepest descent method to obtain the optimal ultrashort laser pulse that injects photons into the cavity and drives the light-matter dynamics to a desired state. Given the feasibility of the proposed scheme, our work paves the way for the development of molecule-assisted sources of light with different photon statistics, for potential applications in quantum metrology in the mid-infrared range.

Edgar Barriga, Millennium Institute for Research in Optics

Scalable realization of quantum Fourier transform interferometers for boson sampling certification

Quantum computational advantage consists of proving that operations carried out on hardware based on quantum mechanical phenomena would allow calculations that no classical computer could perform in a reasonable time. A promising proposal for demonstrating quantum computational advantage is boson sampling. One of the challenges in validating boson sampling lies in demonstrating that the optical device performing this task (boson sampler) is governed by quantum mechanics. This certification problem can be addressed using the quantum Fourier transform interferometer (QFTI) by means of the suppression law. Here, we show how to assemble the QFTI in an optical device in a scalable manner. Using building blocks for the QFTI in small dimensions and the tensor decomposition of the Fourier matrix, we present designs to realize the QFTI for thousands of waveguides, requiring only a few stages of interaction.

Luciano Toro, Universidad de Chile

Thermo-orientation-Induced Complex Structures in Nematic Liquid Crystals

Thermo-orientation describes the preferential alignment of a system induced by a temperature gradient. In nematic liquid crystals, thermal gradients exert an effective torque on the director, competing with elastic forces and surface anchoring. This effect becomes relevant near the nematic–isotropic transition, where reduced elastic constants facilitate reorientation.

Experimentally, we study the nematic liquid crystals 5CB and E7 in homeotropic cells subjected to localized heating. We find that the geometry of the thermal gradient determines the orientational structures: bilateral gradients generate regions with opposite director tilts separated by domain walls, whereas radial gradients promote the formation of vortices.

These observations are complemented by numerical simulations of the temperature field and by a dynamical model derived from the Frank–Oseen energy, including coupling between the director and the thermal gradient. Together, experiment and modeling show that thermal gradients provide a route to controlling nematic orientation and defect formation.

Daniel Uzcategui, Universidad Católica de la Santísima Concepción

Construction of bipartite density matrices with bound entanglement using deep learning tolos

In this work, we develop a method for the generation of bipartite density matrices with bound entanglement using deep learning tools. Our approach consists in first training convolutional autoencoders to distinguish entangled from separable bipartite states for dxd systems with local dimension d ranging from three to seven, which enables identification of bound and free entanglement. Through extensive numerical simulations across various quantum state families, we demonstrate that our model achieves high classification accuracy. Then, we leverage the learned representations to generate novel samples of bound entangled states and certify them using other entanglement criteria.

Stephen Walborn, TBA

Eloísa Pérez, Universidad de Santiago de Chile

Influence of cobalt and nickel metal ions on the optical properties of doped carbon dots for future applications as peroxidase-like nanozymes

This study evaluates the influence of varying proportions of cobalt and nickel ions on the optical properties of metal-doped carbon dots (CDs). The materials were synthesized in situ via a hydrothermal method at 160 °C, utilizing citric acid and ethylenediamine as carbon and nitrogen precursors, respectively, along with cobalt chloride and/or nickel chloride as metal precursors.

The resulting samples were structurally characterized by X-ray diffraction (XRD) and Fourier-transform infrared spectroscopy (FTIR), and optically characterized via UV-Vis and fluorescence spectroscopy. To evaluate the influence of the metal ions on the optical properties and assess their application as peroxidase-like nanozymes, fluorescence was studied in systems combining CDs, a chromophore, and hydrogen peroxide in PBS buffer.

Nikolás Naranjo, Universidad de Chile

Experimentally simple non-classicality detection of Generalized Coherent States

Recently, our group has studied non-classical state generation by generalizing the interaction of a coherent state with a non-linear medium. However, this doesn't change the Poissonian statistics: the resulting state belongs to the family of Generalized Coherent States (GCSs), and its nonclassicality cannot be revealed using standard photon correlation measurements, an alternative to the full state tomography.

We will report on our latest findings on simple methods to certify GCS's non-classicality. We argue that well chosen partial information about the state is sufficient to reveal non-classicality. We've shown that the measurement of the intensity-field correlation function reveals the nonclassicality of GCSs. A witness that relies on the coherence of the state. We now show that a simple photon correlation measurement can also reveal the state's nonclassicality through anti-bunching, considering that the generated state is properly displaced in phase space before measurement. This extends the regime of certifiable non-classical states.

Carlos Pirela, Universidad de los Andes

Experimental evaluation of photon-counting quaternary PolSK for Free-Space Optical Communications

As urban Free-Space Optical (FSO) networks advance, physical layers must ensure both data security and link robustness against atmospheric turbulence. Traditional intensity modulation degrades severely under turbulence-induced scintillation, making Polarization Shift Keying (PolSK)—which encodes data in relative polarization states—a compelling, self-referencing alternative. In this work, we evaluate photon-counting PolSK using a dual Single-Photon Avalanche Diode (SPAD) receiver across simulated and laboratory turbulence channels, focusing on 4-PolSK and scalable M-ary architectures. We show how photon integration timing influences error rates, and reveal why direct-detection nonlinearities degrade performance under standard linear constellation spacing. To overcome these limits, we introduce an equal-probability constellation geometry paired with dynamic decision thresholds that equalize noise margins. Our findings establish a practical, turbulence-resilient architecture for future quantum-limited optical links.

Marcel Clerc, Universidad de Chile – TBA

Andrés Seguel, Universidad de los Andes

Mitigating atmospheric turbulence in free-space single-photon links using transmitter spatial diversity

A two-transmitter, single-receiver setup is used to study the effectiveness of spatial diversity in mitigation of atmospheric-turbulence-induced losses in a single-photon free-space quantum communication channel. The system alternates between transmitters, dynamically selecting the one that best matches current optical path conditions. A simulation framework was developed to evaluate the impact of transmitter separation under different turbulence strengths against a single-transmitter baseline. The experimental implementation features a 515-nm weak coherent state signal collinear with a 660-nm beacon laser, which serves to evaluate path conditions and enable transmitter-receiver synchronization. Progress on the field implementation over a 3-km link—connecting the Cerro Calán Observatory and the Faculty of Engineering at Universidad de los Andes—is discussed, along with preliminary experimental results.

Gastón González, Universidad de Santiago de Chile

THz-to-Optical quantum transduction with Dibenzoterrylene in Anthracene

Single fluorescent organic molecules in molecular crystals are promising platforms for hybrid THz-to-optical quantum interfaces. THz-driven vibrational population is converted into optical photons through a vibronic (Franck-Condon) coupling η combined with the IR/THz dipole activity ξ. The per-mode figure of merit is η· ξ. Isolated dibenzoterrylene (DBT, $C_{38}H_{20}$) is reported as centrosymmetric, mutual exclusion forbids simultaneous Franck-Condon and THz activity (η· ξ = 0), so direct transduction is symmetry-forbidden. Here DBT is embedded in anthracene, where the host can break this symmetry. Ab initio calculations yield distinct DBT structures and insertion geometries, the resulting slight symmetry breaking can lift the exclusion rule and activate Franck-Condon and IR/THz character simultaneously in some modes. From the ab initio parameters we build a Holstein-type vibronic Hamiltonian to model coherent THz and optical driving, a first quantitative framework for transduction in real molecular crystals.

Dario Pérez, Pontificia Universidad Católica de Valparaíso

Direct testing of turbulence statistics with kilohertz complex-field reconstruction

Optical-turbulence theory assumes the complex phase is Gaussian, yet this has rarely been tested directly. Using sixteen Hadamard coded apertures on a micromirror device, we reconstruct amplitude and phase at 1 kHz through controlled turbulence, without any statistical prior. Across 24 conditions, the coherence measured directly differs systematically from its Gaussian prediction, and the gap grows with turbulence strength. The Gaussian model underestimates the coherence radius and its tails, while departures from normality cluster into strongly fluctuating spatiotemporal structures. For quantum optical communications, Gaussian channel models misjudge single-mode fibre coupling and the coherence on which quantum links rely, which calls for statistically faithful channel models.

Alberto Navarrete, Universidad de Santiago de Chile

Engineering Electronic Structure and Crystal Growth in Co/Zn-BTC MOFs towards Photonic Applications

Understanding how crystal chemistry shapes the optical response of metal-organic frameworks (MOFs) is essential for developing materials towards photonic and optoelectronic applications. This work investigates two complementary engineering strategies in Co/Zn-BTC and Co/Zn-BTC-2,2'-bipyridine MOFs to tailor their electronic structure and optical response. The first explores Co/Zn doping and heterometallation with Li, Mg and K as approaches to modify the local coordination environment, influencing metal–ligand electronic transitions and their relationship with band-gap evolution. The second examines shear-controlled laminar flow during the early stages of solvothermal synthesis as a strategy to improve crystallinity and promote larger single-crystal growth, a key parameter for optical studies in crystalline MOFs. Ongoing characterization by PXRD, UV–Vis absorption, photoluminescence and band-gap analysis will be correlated with composition and crystallization conditions to establish composition–crystallization–property relationships.

Romina Abarca, Universidad de Chile

Experimental Observation of BICs in a Giant-Atom-Coupled Lattice

The main feature of a giant atom is the nonlocality of its couplings, allowing them to interact with two distant sites of a lattice. On the other hand, bound states in the continuum (BICs) are localized waves that coexist with radiative modes while remaining confined. In this work, we experimentally demonstrated the excitation of a BIC in a one-dimensional lattice coupled to a giant atom. We propose using photonic molecular links to connect the giant atom to the lattice. The experiments were implemented using the femtosecond laser writing technique, while a Spatial Light Modulator (SLM) was used to excite the system by controlling the amplitude and phase of the input light.

Samir Das, Universidad de Santiago de Chile

Intravibrational coupling enhances Purcell effect

We investigate the relaxation dynamics of a coupled vibrational system in which one vibrational mode is tuned across two other resonant modes while the system is driven by an ultrashort femtosecond laser pulse. We show that Purcell, cross-Purcell, EIT, and anti-Purcell features emerge within a unified framework for an infrared nanocavity. Our results establish intravibrational coupling as a mechanism for controlling vibrational relaxation with infrared light, opening routes toward coherent control schemes with potential applications in quantum chemistry and quantum information processing.

Aldo Delgado, TBA

María Paz Barría, Universidad de Concepción

From Berry to Uhlmann: Optimal Parallel Transport of a Noisy Qubit

Berry's phase captures the geometry of pure states transported along a path, but decoherence turns pure states into mixed ones. Uhlmann's parallel transport extends this geometry by selecting, among all purifications of a mixed-state evolution, the one that moves the least: the horizontal lift, whose speed is set by the Bures metric and the quantum Fisher information. We present an explicit recipe for this lift based on the Uhlmann connection and apply it to a qubit undergoing amplitude damping. We show that the standard Stinespring dilation of this channel is not horizontal, and that the optimal purification follows from it by a single rotation of the environment. This lift implements the channel with the minimal energy dispersion, about 14% below the standard dilation. Finally, we discuss a possible test in polarization–path optical setups and the route towards measuring Uhlmann holonomies.

Nelson Villalba, Universidad de Concepción

Poled-fibre phase modulator for efficient high-dimensional quantum measurements

Efficient detection of quantum states underpins advanced device-independent quantum-information protocols that provide the ultimate level of security for tasks including quantum random number generation and quantum key distribution (QKD). High-dimensional encoding is a natural route to boost the performance of such protocols, offering enhanced noise resilience and higher information capacity, yet their practical implementation remains challenging. A key experimental bottleneck in higher dimensions is the typical need of active modulators for basis selection, which incur substantial optical losses and polarization-sensitive operation. Poled optical fiber phase modulators (PFPMs) are a fiber-native electro-optic technology that naturally addresses these challenges, combining sub-dB insertion loss, intrinsic polarization independence, and direct compatibility with standard telecommunications fiber. Here we report the first use of a PFPM for active quantum-state measurements in a fully fiber-integrated platform. Basis selection in our receiver for four-dimensional qudits is achieved using a single PFPM, substantially simplifying the receiver architecture. As a benchmark, we perform a four-dimensional QKD session and obtain a finite secret-key rate per pulse that, to the best of our knowledge, surpasses all previously reported QKD demonstrations. Our results establish poled-fiber electro-optic modulation as a broadly applicable platform for high-efficiency detection in fiber-integrated quantum information processing.

Priscila Caces, Universidad de Santiago de Chile

CD-Ni/ZIF-8-Based Enzymatic Biosensor for Glucose Detection

Glucose detection represents a challenge, as the methods currently used can be costly and, in some cases, invasive. In this context, the present work aims to develop an enzymatic fluorescent sensor capable of detecting glucose in a sensitive, specific, and non-invasive manner. To this end, a composite material was developed based on nickel-doped carbon dots (CD-Ni) and Zeolitic Imidazolate Framework-8 (ZIF-8), a type of Metal-Organic Framework (MOF). The results obtained demonstrated that the developed material exhibits sensitivity to the presence of glucose, demonstrating its ability to respond to the detection of the target analyte. Furthermore, the sensor showed specific behavior toward glucose when evaluated in the presence of different potentially interfering substances. These results suggest that the CD-Ni/ZIF-8 composite has potential for the development of glucose detection systems.

Mauricio Morel, Universidad de Atacama

Light- and heat-actuated films based on chiral nematic liquid crystals

Molecular orientation in chiral nematic films dictates their optical, thermal, and mechanical properties. Although photoisomerizable azo compounds facilitate macroscale reorientation, how the exact ratio of chiral to azo molecules regulates this process within polymer matrices remains poorly understood.
This study investigates how varying component concentrations influence the efficiency and stability of light- or heat-induced reorientation. Methodologically, the commercial liquid crystal mixture E7 was doped with a synthesized liquid crystal and a chiral molecule. This system was combined with methyl red (0.5%–1.0%) and varying proportions of a commercial monomer to synthesize films via photopolymerization.

Analysis under external stimulation demonstrated that the films exhibit preferential actuation (either optical or thermal) dictated by the specific azo-compound concentration. Furthermore, the system manifested distinct hysteresis when subjected to repeated stimulation cycles.

Lucas Hernández, Universidad de Chile

Optical vortex generation on photonic chips

Optical vortices, or beams with orbital angular momentum, are the simplest solution to the Hemholtz equation with non-trivial topology, thanks to its phase singularity. Exploiting such phase structure, it's possible to design waveguide lattices with complex coupling, and arbitrary effective magnetic fields.

Vortex generation is usually performed using an SLM, but requires a more elaborate experimental setup. Generating methods using curved waveguides have also been proposed, but this requires a more complex fabrication process and induces losses.
In this work we propose, simulate, and experimentally implement a generation method consisting of only 3 masks of straight waveguides. This method generates vortices with charge ±1, depending on the input, which can be chosen continuously without modifying the generator. It also serves as a visual interferometer, allowing the observation of small phase differences.

This work provides an exchange between simplicity and flexibility, while also opening up new integrated experimental methods.

Belén Orellana, Universidad de Santiago de Chile

Stoichiometric Control of Structure and Charge Transport in Fe-MOF-Based Materials

Metal-organic frameworks (MOFs) are promising electrode materials for supercapacitors owing to their tunable coordination chemistry, porous architectures, and accessible redox–active sites. However, their practical electrochemical performance is often limited by poor charge transport, restricted ion accessibility, and structural instability. In this work, Fe-chloranilate coordination materials were synthesized by a hydrothermal route using a redox-active chloranilate ligand, and the effect of the metal-to-ligand molar ratio on their structural evolution and electrochemical behavior was systematically investigated. Three (Fe)(ligand) compositions, namely (6)(3), (3)(3), and (3)(6), were prepared and characterized by powder X-ray diffraction, FTIR, XPS, SEM-EDX, AFM, and electrochemical impedance spectroscopy. The results reveal that small changes in the synthesis stoichiometry strongly affect phase formation, morphology, and interfacial resistance. The strong influence of the Fe-to-ligand ratio on the structural, morphological, and charge-transport properties of these materials suggests that their coordination chemistry could be exploited to tune their electronic and, potentially, optical response, opening opportunities for optical sensing and optoelectronic applications.

Bastián Real, Universidad Técnica Federico Santa María

Directional control through coherent driving in Hermitian and non-Hermitian photonic lattices

Controlling where light travels is key to optical information processing. In this talk, I will show how resonant lasers with tailored amplitudes and phases interfere with the modes of photonic lattices to steer light at will. In two-dimensional lattices, tuning the drives to the energy of band saddle points yields quasi-one-dimensional, unidirectional propagation, as well as localized states of controllable extent. Specifically, in a one-dimensional lattice of semiconductor micropillars, we experimentally demonstrated optical switching, directional transport with up to 92% of the light sent to one side. Finally, I will present recent results on non-Hermitian 1D lattices. With alternating gain and loss, directionality becomes sharper but is lost at the exceptional point, where light localizes. Adding non-reciprocal couplings restores directional propagation within an energy band whose width grows linearly with the strength of the non-reciprocal parameter.

Santiago Ocampo, Universidad de Chile

Dancing Hula Hoop Poincare Beams

When light interacts with birefringent materials, it experiences a polarization-dependent phase shift. Inhomogeneous birefringent media can therefore generate spatially structured light, where each point in space corresponds to a different polarization. These solutions, known as Poincaré beams, are naturally generated using nematic liquid crystals. Here, topological vortex defects interact with light to map these states onto the Poincaré sphere. However, at high voltages, the cell's finite thickness forces the vortex to twist longitudinally. This causes the structured solution to acquire a phase shift along the sphere's equator, a behavior coined as the Hula-Hoop Poincaré beam. In this work, we show the dependence of this twist on the applied voltage and propose a theoretical solution for the director field that explains it. The significance resides in the theoretical correction to liquid crystal theory and future applications in optical engineering.

Tomás Veas, Universidad de Concepción

Phase Estimation of Entangled Light Using a Quantum Mirror

This work proposes a protocol to determine the relative phase of a two-mode entangled state of light, based on a single quantum mirror. This device enables a considerably simpler method of inducing entanglement in a quantum state of light, and also allows quantum-state inteferometry through the measurement of a single ancillary atom, instead of measurements over the optical modes. We have demonstrated that a wide variety of exotic quantum light states provide metrological advantage over the shot noise limit (standard quantum limit). Under the correct conditions, these states saturate the Heisenberg limit for the phase estimation.

Sofía Pinto, Universidad de Chile

Multimode Photonic Lattices for Bound States in the Continuum

Photonic lattices have become a powerful platform for investigating transport phenomena in quantum systems, owing to the mathematical analogy between the Schrödinger equation and the paraxial wave equation. In this work, we implement this correspondence in a tight-binding lattice supporting a bound state in the continuum (BIC), a spatially localized state with an energy lying within the system's continuous radiation spectrum.

The waveguides are fabricated in borosilicate substrates using femtosecond laser direct writing. By combining s and p modes at specific lattice sites, next-nearest-neighbor coupling can be suppressed, enabling a more faithful experimental realization of the theoretical model. To investigate scattering in the central lattice, an additional edge waveguide is excited with a supercontinuum laser, generating an effective plane-wave excitation with a well-defined momentum distribution and allowing us to probe the scattering properties of the system.

Álvaro Alarcón, Universidad del Biobío

From the Periphery: Rethinking Quantum Technologies as Scientific, Social and Cultural Experiments

Quantum technologies are usually presented through a promise: unconditional security, computational supremacy, industrial disruption. Yet those of us who work in the lab know that an experiment is much more than its promise. Noise, imperfections, the experimenter's skills, competitive funding, patents and social expectations are part of the same assemblage that produces photons, interferometers and networks.

In this talk, we propose looking at quantum technologies as infrastructures that generate not only knowledge but also new languages, practices, institutions and ways of inhabiting the world. Drawing on the experience of the Quantum UBB group (fiber interferometers built by undergraduate students, low-cost hardware and STS-based teaching), we will discuss why philosophy, anthropology and the social sciences are not a complement to quantum development but a constitutive part of it. We argue that perspectives now considered peripheral must take a central role in building quantum technologies and in understanding what they really are.

Adrián Rubio, Universidad de Santiago de Chile

Oscillators under frequency noise: towards applications and limitations

In Physics, oscillators show as fundamental bricks over which both theory and experiments build on. Their relatively simple physical features result in an advantage given the possibilities for an accurate description and precise control. In this sense, understanding how oscillators are affected by different contexts and the influence of different external phenomena, such as dissipation, noise and energy injection of different kinds, remains crucial for achieving novel avenues in research and technology. In this talk, we specifically focus on how frequency noise affects harmonic systems. The energy injection by frequency noise allows to enhance modifications in the oscillator's observable physical quantities at the steady state due to its interaction with structured environments consisting in additional oscillators and thermal baths. Finally, we will comment on the possibility of employing these phenomena for thermal rectification of heat currents and observing zero-point fluctuations in baths even beyond the low-temperature regime.

Rubén Fritz, MAQI Spa.

A Molecular-Based Approach Toward High-Throughput Screening of Metal–Organic Frameworks for Second-Harmonic Generation

Metal–organic frameworks (MOFs) are versatile materials whose modular structure, built from metal nodes and organic linkers, allows their physical, chemical and biological properties to be tuned by design. This tunability makes them attractive for nonlinear optics, a field still dominated by a handful of inorganic crystals, such as KDP, KTP and BBO. Indeed, several MOFs have shown promising second-harmonic generation (SHG) responses both theoretically and experimentally, but exploring entire MOF databases experimentally or with ab initio methods remains prohibitive. To address this, we implement a molecular physics approach to predict the magnitude of the SHG response in a selected set of MOFs. We benchmark the predictions against state-of-the-art density functional theory (DFT) calculations and assess the accuracy and computational cost of the method. Finally, we discuss whether such approaches can enable high-throughput screening to identify new MOFs with strong nonlinear optical properties.

Polette Parra, Universidad de Chile

Generation of line states in diamond and graphene lattices

The generation of compact states in diamond and graphene rings has been demonstrated. When additional rings are joined at the nodes to form a ribbon, the system exhibits flat bands; however, increasing the dimensionality causes the system to lose them. This work demonstrates the generation of "column states" in graphene and diamond lattices, showing that they persist regardless of the lattice size. Furthermore, in the case of diamond, the orthogonality of these column states with "row states" is demonstrated, giving rise to 2^n combinations for an n x n lattice.

Felipe Herrera, Universidad de Santiago de Chile, TBA

Hans Mättig, Universidad de Concepción

Rigorous quantum state tomography for distributed quantum computing

Distributed quantum computing offers a promising approach to scaling quantum devices by networking multiple quantum processors. We present a quantum state tomography protocol tailored for distributed quantum computers that avoids assuming remote entanglement as a primitive resource. We create a tomographic protocol that uses 2-designs measurements and we also create a classical shadows method that allow us to obtain properties of the system. We derive rigorous, non-asymptotic trace-norm error bounds for the state estimator, with explicit exponential dependence on the number of nodes. In addition, we estimate a bound for the shadow norm and the variance of the properties, which scales with the number of non trivial nodes present in the distributed environment. Numerical simulations for systems of up to seven qubits distributed across several devices validate the theoretical error bounds of the 2-design method.

Manuel Díaz, Universidad de Chile

Light Induced Polarization Transition from Scalar to Full Poincaré Beam in Optical Valves

We demonstrate a light-controlled mechanism for the generation of vector Poincaré-type beams mediated by localized vortex defects in liquid crystal light valves. A Gaussian beam induces a localized vortexlike molecular reorientation pattern, with an amplitude governed by the incident intensity profile. This self-induced anisotropic structure transforms an initially scalar beam into a spatially structured polarization field. To capture the underlying nonlinear light-matter interaction, we derive a minimal dynamical model from first principles and establish its direct connection with Poincaré-beam generation. Combining experiments and numerical simulations, we show that the polarization texture can be continuously tuned through the input intensity and the applied voltage frequency, allowing a controlled transition from a homogeneous scalar beam to a full Poincaré beam. We further demonstrate that this mechanism extends naturally to programmable lattices of Poincaré beams, emphasizing the versatility of liquid crystal light valves as reconfigurable platforms for structured-light generation.

Jean Cortés, Universidad de Concepción

Stochastic Optimization of Nonlocality Inequalities in Bell Networks

We present a self-guided method for detecting nonlocal correlations in the triangle network, where three independent sources connect three measurement nodes. The method optimizes the nodes' joint measurements to seek violations of the Wagon-Wheel inequality, without assuming a target probability distribution or predetermined measurement settings. This provides a way to explore nonlocality across different distributed quantum states.

Numerical results show that the protocol finds violations with maximally entangled sources and remains informative when entanglement or noise levels vary. We also investigate how the method performs when probabilities must be estimated from finite measurement data. By incorporating Classical Shadows, the optimization can reuse measurement data across candidate settings. The results reveal how statistical uncertainty affects the search and indicate the measurement resources needed to obtain reliable violations. Together, these findings support self-guided optimization as a practical approach to studying nonlocality in quantum networks.

Diego Guzmán, Universidad de Chile

Observation of Aharonov-Bohm caging and Fano resonances in optical lattices

In this talk, we will present the experimental realization of two phenomena relevant to condensed matter physics: Aharonov-Bohm caging and Fano resonances. Using the femtosecond laser direct writing technique, we fabricated optical lattices and obtained experimental images revealing the characteristic signatures of both phenomena.

Osmar Aravena, Universidad de Santiago de Chile

Optical Response in Planar Waveguides for Enhanced Methane Gas Sensing

Methane detection is critical for environmental monitoring and mitigating industrial explosion risks. Optical planar waveguides present an attractive platform for gas sensing due to their compact footprint, high electromagnetic immunity, and strong evanescent field interaction. In this work, we propose and analyze a planar waveguide methane sensor operating through gas-pressure-induced variations in the optical parameters of the guided mode.By modeling the pressure dependence of the surrounding gas medium and its coupling to the evanescent field, we demonstrate how operational pressure shifts modulate the effective refractive index and propagation metrics within the waveguide core.Enhancements in sensing sensitivity are achieved by identifying optimal parameters for higher-order mode excitation via a Gaussian beam.

Diego Hidalgo, Pontificia Universidad Católica de Chile

Development of a Modular and Quantitatively Characterized Raman/SERS Platform

Raman and surface-enhanced Raman scattering (SERS) experiments in our laboratory are evolving from preliminary demonstrations toward a quantitatively characterized, modular instrumentation platform. Our Au-based SERS chip results have already been reported in a preprint, and final measurements are now being acquired to complete the statistical and spectroscopic validation for publication. In parallel, an OpenFlexure Delta Stage is being characterized using knife-edge measurements to quantify its XYZ positioning performance and assess its suitability for Raman mapping and future imaging. The same open-source platform is also being used for measurements of gramicidin thin films, compared with powder samples, using near-infrared continuous-wave excitation derived from a home-built Ti laser. Together, these developments demonstrate the consolidation of a predominantly in-house Raman/SERS architecture, now supported by quantitative metrics, reproducible measurements, and increasingly diverse applications.

POSTERS

Santiago Acevedo, Universidad de Concepción

Quantum Gravity Certification via Monogamy Inequalities

The Bose-Marletto-Vedral model tries to measure the gravitational entanglement between two macroscopic masses, but this implies serious experimental difficulties, such as the need for macroscopic quantum parameters or issues associated to non-gravitational interactions.

To overcome these experimental challenges we switch from a bipartite system to a tripartite one, and use the Coffman-Kundu-Wootters monogamy inequalities; which states that entanglement is a monogamist resource: if a qubit A is maximally entangled with B, then it cannot be entangled with another C, and any decrease in the observable entanglement between A and B implies that the entanglement has migrated to C.

Thus, we start from a maximally entangled state and then migrate the entanglement to a system G by gravitational interaction, and demonstrate how the monogamy inequalities allow for the certification of quantum traces of the spacetime metric in the weak-field regime, without requiring perfect isolation or direct measurements of the gravitational field.

Mauro Cid, Universidad de Concepción

Quantum State Tomography

The frontier of quantum computing is continually expanding, making the estimation of quantum states a fundamental task. Quantum state tomography reconstructs a state from a large number of measurements using different statistical methods. In this work, through computational simulations, two estimators were compared: direct linear regression and maximum likelihood estimation (MLE). The quality of the estimates was quantified using fidelity between the original and estimated state.
Linear regression does not always yield valid quantum states; MLE, under the imposed constraints, does reconstruct physical states. Results show that MLE outperforms linear regression when estimating a qubit, achieving acceptable fidelity values with fewer repetitions. This highlights the importance of choosing optimal estimators according to the desired precision.

Adheris Contreras, Universidad de Concepción

Inline inter-core phase control in multi-core fiber with a commercial polarization controller

Inter-core phase control is a prerequisite for interferometry in multi-core fiber (MCF), but no compact inline element for this purpose is presently available, and existing approaches require fanout to single-mode fiber or free-space optics. We show that a commercial inline polarization controller, can be operated as an inline inter-core phase shifter. Two-core interferometry on a jacketed four-core patchcord gives a reversible inter-core phase of 0.95 ± 0.02 rad from a single device, driven through repeated load-unload cycles. A diametrically compressed disk model is used to predict the operating condition: with the input polarization aligned to a principal stress axis the device imparts a relative phase between cores without rotating the polarization state of either, which we verify by polarimetry on all cores. The result is a practical, inline route to inter-core phase control and stabilization in all-MCF interferometers, requiring neither fanout to single-mode fiber nor free-space optics.

Marco Durán, Universidad de Santiago de Chile

Optical Properties of ZIF-8 Composites Incorporating NCDs, Ni-NCDs, and AgNPs

This work studies the improvement in optical properties of the zeolitic imidazole framework 8 (ZIF-8) when nitrogen-carbon dots (NCD), NCD doped with nickel (NCD-Ni) and silver nano-particles (AgNps) are incorporated into its structure. ZIF-8 is a stable and biocompatible material, while NCD are known for their excellent optical properties, these characteristics make a composite material from ZIF-8 and NCD a novel and ideal candidate for development of sensing devices. Moreover, metals like Ni and Ag can improve catalytic performance of other materials, thus helping in sensing applications. 6 materials were analyzed via fluorescence intensity measurements and Ultraviolet-Visible analysis; ZIF-8, ZIF-8@NCD, ZIF-8@NCD-Ni, ZIF-8@AgNps, ZIF-8@AgNps@NCD-Ni, ZIF-8@NCD-Ni@AgNps. This work notes the influence of NCD and AgNPs as well as the effect of the incorporation order.

Víctor Gondret, Universidad de Chile

Experimental generation and characterization of non-classical light

The generation of non-classical states of light is one of the key challenges in quantum optics and quantum technologies. In recent months, we have developed and characterized a bright two-mode squeezed-state source generated through the interaction of light with a hot rubidium vapor [1], that we compare with a microscopic [2] and macroscopic model [3]. In our experiment, an intense pump field parametrically drives the χ(3) nonlinear medium, where two pump photons are converted into probe and conjugate photons while conserving energy and momentum. We use an additional beam, referred to as the seed, which stimulates the amplification process and enables operation in the bright-field regime. Building upon these tools, we propose to encode hidden information in the two modes of bright-squeezed state.

[1]. C. Cerda, tesis de Magíster (2026).
[2] H.M. Flores, arXiv 2512:15051 (2025); Basilio et al, in preparation.
[3]. F. Ramírez et al., arXiv 2606:21692 (2026)

Nayda Guerrero, Universidad de Concepción

High-Dimensional Quantum Key Distribution via Full Core-Mode Encoding over Deployed Multicore Fibers

Quantum key distribution (QKD) provides information-theoretic security rooted in quantum physics, while high-dimensional (HD) encoding increases both noise tolerance and secret-key yield. Field demonstrations over deployed multicore fibers (MCF) have so far relied on a hybrid qudit encoding strategy that combines two path with the time-bin photonic degree of freedom, rather than exploiting the full set of available core modes. Although practical, this current approach incurs intrinsic efficiency penalties that grow with dimension.

In this work we implement a d=4 HD-QKD protocol based on pure core-mode encoding over a deployed MCF network across the Universidad de Concepción campus, operating under continuous and uncontrolled environmental perturbations. The reconstructed states using quantum tomography has a fidelity F=0.96±0.01 and a secret key rate of R=5.36 x10^{-3} bits per pulse. Our results thus establish core-mode encoding as a hardware-efficient and field-viable framework for high-dimensional quantum-secure communications.

Maryam Hamed Conejeros, Universidad de Chile

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To overcome these experimental challenges we switch from a bipartite system to a tripartite one, and use the Coffman-Kundu-Wootters monogamy inequalities; which states that entanglement is a monogamist resource: if a qubit A is maximally entangled with B, then it cannot be entangled with another C, and any decrease in the observable entanglement between A and B implies that the entanglement has migrated to C.

Thus, we start from a maximally entangled state and then migrate the entanglement to a system G by gravitational interaction, and demonstrate how the monogamy inequalities allow for the certification of quantum traces of the spacetime metric in the weak-field regime, without requiring perfect isolation or direct measurements of the gravitational field.

María José Herboza, Universidad de Concepción

Bell test to distinguish Post-Newtonian metric theories

We present an experimental proposal for discriminating metric theories of gravity by estimating Post-Newtonian (PN) parameters in quantum systems. This work analyzes gravitationally induced phase shifts to constrain deviations from the General Relativity values of the PPN parameters (gamma=1, beta=1). The phase shifts are probed by measuring their effect on particle entanglement, as quantified by the CHSH inequality. We analyze a Hugged-type interferometer in which one of its constituent Mach-Zehnder interferometers can rotate around the lower horizontal axis. We discuss the modeling and present preliminary results for the estimation of $\delta\gamma$ and $\delta\beta$. Finally, we discuss future prospects for extending this method to estimate the frame-dragging parameter (alpha), e.g., using a quantum Sagnac interferometer.

Carlos Jara, Universidad de Concepción

Finite-Key Analysis of High-Dimensional Quantum Key Distribution

High-dimensional quantum key distribution (HD-QKD) offers increased information capacity per detection event, but demonstrating its practical performance requires rigorous security bounds that account for finite data and statistical fluctuations. This work presents the finite-key analysis of four-dimensional (d = 4) decoy-state QKD implementations based on core-mode encoding in multicore fibers.

The analysis follows the vacuum-plus-weak-decoy framework of Zhang *et al.* (2017), adapted to high-dimensional protocols. Inverse Chernoff bounds are used to estimate the vacuum and single-photon contributions from experimentally measured detection statistics, while random sampling provides an upper bound on the single-photon error rate. These bounds are combined with the generalized Shannon entropy to obtain composable secret-key lengths under finite-size security constraints, without relying on Gaussian approximations for statistical fluctuations.

Giovanni Marín, Universidad de Santiago de Chile

Optoelectronic Properties of $CuInTe_2$ under UV, Visible and Near-Infrared Illumination: From Photoconductivity to Photodetection in the CdS/$CuInTe_2$ Heterojunction.

Authors: Giovanni Marín Lobo and Dinesh Pratap Singh

$CuInTe_2$ is a ternary I–III–VI_2 chalcopyrite semiconductor with predominantly p-type conductivity and a direct optical band gap of approximately 0.94–1.14 eV, making it promising for visible and near-infrared optoelectronic applications. In this work, the photoelectric response of $CuInTe_2$ is investigated as a function of excitation wavelength and optical power density. Photoconductive measurements are performed using In and Ag contacts under UV (375 nm), visible, and near-infrared (1200 nm) illumination, including intermediate wavelengths from 370 to 850 nm. The wavelength-dependent photocurrent is analyzed to assess spectral sensitivity and photocarrier transport. Additionally, an n-CdS/p-$CuInTe_2$ heterojunction is investigated to evaluate its photovoltaic response under different illumination conditions. Finally, transient measurements under repeated illumination and dark cycles are performed to characterize the dynamic photodetection response. These results provide insight into the spectral, intensity-dependent, and temporal optoelectronic properties of $CuInTe_2$ for potential photodetection applications.

Francisco Molina, Universidad de Chile

Phase separation and pattern formation in cholesteric Liquid Crystal

Liquid crystals are materials capable of forming a wide variety of spatial patterns due to the interaction between their molecular organization and external or internal conditions. In the chiral liquid crystal mixture E7FEOS12, different domains can be observed, suggesting phase separation caused by local variations in concentration. In this work, we study this phase separation using numerical methods. The concentration dynamics are described through the Cahn–Hilliard model, while the formation of chiral patterns is modeled using a chiral anisotropic Ginzburg–Landau equation. Both models are coupled by assuming that the local chirality depends on concentration. Numerical simulations were used to explore how concentration evolves in time and how phase separation modifies the spatial organization of the system. This approach provides a simple framework to study the connection between local concentration variations, phase separation, and pattern formation in chiral liquid crystals.

Matías Morales, Universidad de Concepción

Sub-picosecond inter-core skew characterization in multicore fibers via Hong–Ou–Mandel interference

Inter-core skew (ICS), the differential group delay between cores of a multicore fiber, is a critical parameter for classical space-division multiplexed communications and quantum photonic networks. We present a high-precision measurement of ICS in four-core fibers using two-photon Hong–Ou–Mandel (HOM) interference in a fiber-integrated 4 × 4 multiport beam splitter, using spontaneous parametric down-conversion (SPDC) and a weak-coherent-state source. By extracting the center position of HOM interference dips and peaks across all core-pair combinations, we obtain individual ICS values. This multi-length characterization of RMS ICS, finds results consistent with the expected $\sqrt{L}$ stochastic random-walk scaling, $\sigma_\tau(L) = \kappa\sqrt{L}+c$ with intrinsic coefficient $\kappa = (39.80\pm0.003)$ ps/$\sqrt{}$km. The demonstrated ±0.11 ps precision for the two-photon interference represents a ∼180-fold improvement over C-OTDR. The SPDC source provides this precision on laboratory-testbed fibers, while the WCS source trades precision for loss tolerance, extending the measurement to a 1300 m field-deployed fiber.

Luis Núñez, Universidad de Santiago de Chile

Green functions for surface with non-negligible curvature

The Casimir effect is a quantum phenomenon associated with the zero-point fluctuations of quantum fields, the energy of which is modified by the presence of boundaries or boundary conditions. In its simplest configuration—involving two ideal, parallel, conducting plates at zero temperature—an attractive force arises between them. However, its characteristics vary depending on geometry, boundary conditions, material properties, and temperature. The study of this effect has gained particular importance in contexts where the Casimir force can influence the operation of MEMS and NEMS devices. Green's functions serve as a tool for studying these interactions, enabling the incorporation of boundary conditions and the analysis of field response in complex geometries. This work examines the construction of Green's functions for calculating the out-of-equilibrium Casimir force between a point-like object and a surface with non-negligible curvature. In particular, we investigate the non-trivial geometry of a parabolic cylinder.

Matías Pérez, Universidad de Santiago de Chile

Green Synthesis Optimization and Optical Properties of Carbon Dots Derived from Maqui Fruit (Aristotelia chilensis)

Carbon dots (CDs) are carbon-based nanomaterials with interesting optical properties, whose development can be directed towards more sustainable synthesis methods using natural and renewable sources. In this work, a green synthesis route for CDs was developed using freeze-dried maqui fruit (Aristotelia chilensis) as a carbon precursor through a hydrothermal process. To optimize the material production, the effects of the maqui precursor amount and synthesis temperature were evaluated. Additionally, modifications to the dialysis process were implemented to improve the removal of impurities from the synthesized material. The obtained CDs were optically characterized by fluorescence and UV-Vis spectroscopy to evaluate their optical properties. The combined evaluation of synthesis, purification, and optical parameters allowed the establishment of suitable conditions for obtaining maqui-derived CDs. These results highlight the potential of A. chilensis as a natural and renewable precursor for the sustainable development of carbon-based nanomaterials.

Mauro Riquelme, Universidad de Chile

Fractional Topological Defects in a Self-Coupled Liquid Crystal Light Valve

Liquid crystals combine fluidity with orientational order, enabling rich topological phenomena and applications in photonics. In this work, we study a homeotropic nematic liquid crystal light valve (LCLV) subjected to optical feedback via cross-polarization. This feedback modifies the orientational transition from subcritical to transcritical, creating a robust bistable regime where reoriented and non-reoriented states coexist. Within this regime, we observe spontaneous appearance, propagation, and annihilation of matter vortices, as well as front propagation. We derive an amplitude model based on a generalized Ginzburg-Landau equation that, despite arising from a rotationally symmetric system, lacks that symmetry due to self-coupling. The model captures the observed vortex dynamics and supports fractional topological defects of charge 1/2 in the nematic director—solutions rarely accessible to standard amplitude descriptions. Preliminary interferometric measurements provide evidence consistent with these fractional structures.

Cristóbal Riveros, Universidad de Chile

Oscillatory dynamics induced by delayed optical feedback in a liquid crystal valve

In any physical system, a minimum time delay is associated with the interaction between its components, determining the temporal scale at which information is transmitted. When this delay is not negligible, its presence leads to the formulation of dynamical equations with memory, yielding solutions with qualitatively different properties. Here, we present an experimental setup based on a liquid crystal light valve with optical feedback, incorporating a second time-delayed feedback loop to evaluate its effect on the system's dynamical evolution. We identify an Andronov-Hopf bifurcation that generates local oscillations, alongside the emergence of spiral waves when investigating the spatially extended system. A valid model near the bistability zone was established, its corresponding amplitude equation derived, and experimental results compared against numerical simulations of the model. This work establishes a controllable methodology for studying the effects of time delay in nonlinear systems.

Carlos Sáez, Universidad de Chile

Thermal gradient driven orientation in nematic liquid crystals

Nematic liquid crystals are anisotropic mesophases having crystal (nematic) and fluid-like (liquid) physical properties at once, with nanometric rod-shaped molecules forming local average orientated clusters. This orientation can be changed by electric, magnetic, and electromagnetic fields, being the basis of various technological applications, such as displays, sensors, light modulators, etc. We studied the molecular reorientation of nematic liquid crystals confined in glass cells, inducing thermal gradients, observing by cross-polarized light microscopy. We observed the emergence of topological defects during the transition from the liquid state to the nematic state. In addition, the temperature profile and thermal gradient were numerically estimated by solving the heat equation, and the theoretical analysis was carried out by the minimization of Frank–Oseen free energy. We obtained a preliminary scalar model that describes a one-dimensional reorientation, with parameters that depend on known physical constants. The experimental findings are in agreement with numerical simulations.

Mauricio Sánchez, Universidad de Concepción

Assessing the Wigner Phase as a Tool for Post-Newtonian Parameter Estimation

In the context of classical theories of gravity, there are several that describe gravitational phenomena. Among these, the most widely recognized is the General Theory of Relativity. Each of them predicts certain behaviors of nature that we have been able to measure with increasing precision, allowing us to better understand which theory correctly describes nature. To compare these theories, the parameterized post-Newtonian formalism is introduced. In this formalism, there are several parameters that quantify different characteristics related to gravitational behavior. Through an experimental proposal within the context of quantum theory, using interferometry, we propose a scheme to measure some of the post-Newtonian parameters using the gravity-induced variation of light polarization, know as Wigner Phase, measured from a geodesic interferometer. These results will allow us, through a quantum protocol, to discriminate between classical theories of gravity.

Fernando Vera, Universidad de Concepción

Absorption-Free Imaging with Single-Photon Fourier Optics

We introduce Absorption-Free Imaging (AFI), a continuous-space extension of interaction-free measurement in which the transverse field replaces the discrete path register of conventional IFM. Using Fourier propagation and nulling of the no-object reference field, we show that dark-port detections measure how the object's complex transmission varies across the illuminated region, while the camera image maps local deviations from its illumination-weighted average transmission.

For opaque objects, AFI recovers the Elitzur–Vaidman interaction-free probabilities and the single-pass 50% efficiency bound; more generally, the multipath IFM equations converge to the AFI formulation in the continuous-space limit. The same operator structure also establishes a direct connection to coherent dark-field imaging. Experimentally, a coherent laser-SLM implementation tests these linear-optical predictions rather than single-photon statistics.

The broader motivation is to determine how much spatial information can ultimately be extracted per absorption or damaging event, and explore viability for label-free imaging in photosensitive materials and biological samples.